

SITHASSEMBLY // CONTROLLED REFERENCE

External Stack Evaluation Dossier

Current-state assessment, scorecard, and bounded proof-of-concept sequence.

EDITION	2026-09-06
STATUS	Architecture review snapshot dated 2026-09-04
SOURCE	docs/architecture/external-stack-evaluation.md docs/architecture/external-stack-scorecard.md docs/architecture/poc-plan.md
CONTROL	Repository-tracked English source

External Components: Current-State Review

Date: 2026-09-04. Printable snapshot: docs/External-Stack-Evaluation-Dossier-2026-09-04.pdf.

This assessment covers only the current local workspace. It does not adopt software, start services, or replace project-specific logic. Verify every candidate's current release, license, security advisories, and compatibility against primary sources immediately before a proof of concept.

Existing Architecture Map

Area	Existing anchor	Assessment
HTTP and UI	app.py, src/server.py, web/	Local server with API and casework UI. No external gateway is required.
Commands	src/SithAssembly/CommandDeck/command_engine.py, SithAssembly.Runtime.py	Allowlisted local commands against the case store, not a general shell or tool proxy.
Registry and modules	config/module_registry.json, src/module_runtime.py, src/assembly_manifest.py	Explicit src.SithAssembly.* declarations, controlled loading, and a readable manifest.
Coordination	config/agent_registry.json, src/SithAssembly/Conclave/, src/SithAssembly/AssemblyCore/agent_controller.py	Local topic and capability model; no external message broker.
Data and graph	src/database.py, src/repository.py, src/SithAssembly/SpectreNet/, src/SithAssembly/GhostCluster/	SQLite, evidence-bound edges, local groups, and centrality.
Analysis	src/SithAssembly/SignalForge/, src/SithAssembly/GlyphWatch/, src/SithAssembly/MindForge/	Optional local adapters; no silent model downloads or external calls.
Evidence and exports	src/SithAssembly/CipherLedger/, src/SithAssembly/EvidenceVault/, src/SithAssembly/BlackArchive/	Fingerprints, opt-in encrypted packages, and JSON/PDF exports.
Telemetry	src/runtime_logging.py, src/SithAssembly/ForgeProbe/runtime_doctor.py	Local redacted JSONL logs and read-only runtime diagnostics.
Deployment	config/deployment.local.json, src/SithAssembly/Citadel/deployment_preflight.py, deploy/	Inactive preparation topology; no cluster is running.

Candidates and Integration Boundaries

Candidate	Status	Justified integration boundary
IBM ContextForge	Defer	<code>src/module_runtime.py</code> and <code>config/module_registry.json</code> already solve local registry needs. Benchmark only when multiple external MCP/A2A services exist.
agentgateway	Defer	There is no external MCP/A2A or gRPC traffic. A second gateway beside <code>src/server.py</code> would duplicate structure.
NATS JetStream	Plan a PoC	Fits future <code>config/agent_registry.json</code> topics. One event must prove idempotency, replay, backpressure, and failure handling.
Temporal	Plan a PoC	Relevant only for an existing long-running, resumable job. The current request model does not need a workflow server.
Open Policy Agent	Priority: small PoC	<code>config/agent_registry.json</code> contains roles, approvals, and capabilities. A central policy input is valuable only when it fails closed.
OpenLineage and Rekor	Reuse concepts	<code>src/runtime_logging.py</code> , <code>src/SithAssembly/CipherLedger/</code> , and <code>src/SithAssembly/EvidenceVault/</code> are the basis. Specify an internal run/artifact envelope first.
OpenCTI	Reuse data-model patterns	Deduplication, confidence, source priority, and explicitly inferred edges are useful. Do not adopt the full OpenCTI stack.
QUT Coordination Network Toolkit	Evaluate methodology	Test platform-neutral coordination features against <code>src/pattern_engine.py</code> and <code>src/SithAssembly/SignalForge/</code> . Results remain review-required pattern candidates.
OSINTGraph	Reference, not core	Compare import, graph, and UI ideas in isolation. Do not couple to fragile access methods.
OpenTelemetry and AgentOps	Plan OTel-first	<code>src/runtime_logging.py</code> is a local basis. Define a trace contract before an exportable adapter.
Firecracker, gVisor, Microsandbox	Evaluate later on server	No gain on the local Windows workstation. Evaluate untrusted workloads only with risk classes and network boundaries.

Decision

The next useful change is not a framework import. Prioritize three small, measurable contracts: a policy input, event envelope, and provenance envelope. They keep the local architecture compatible with OPA, NATS/JetStream, and OTel without requiring any of those services. The command core and evidence-bound casework remain central.

External Components: Scorecard

Date: 2026-09-04. Scores measure fit with the current codebase, not the quality or security of a third-party project. Scale: 0 unsuitable, 1 weak, 2 limited, 3 usable, 4 strong, 5 very strong.

Weights: architecture fit 20%, eliminated project complexity 15%, operational effort 10%, scaling/backpressure 10%, observability/debugging 10%, replaceability 10%, data/policy control 10%, maturity/maintenance 5%, license/self-hosting 5%, and migration/rollback 5%. Leave unverifiable criteria open instead of guessing.

Candidate	Preliminary score	Category	Repository evidence	Condition before a PoC
OPA	3.8	Plan a PoC	config/agent_registry.json has approvals and capabilities but no central policy contract.	Verify the current license, release, and fail-closed Python integration.
OpenLineage/Rekor concepts	3.6	Reuse concepts	src/SithAssembly/CipherLedger/, src/SithAssembly/EvidenceVault/, and src/runtime_logging.py contain hash and log anchors.	Define a private provenance format; do not export sensitive evidence.
NATS JetStream	3.4	Plan a PoC	config/agent_registry.json already models publishes/subscribes.	An event envelope plus idempotency and replay test must pass.
OpenTelemetry	3.3	Plan a PoC	src/runtime_logging.py redacts locally and src/runtime_doctor.py provides diagnostics.	Define trace and redaction contracts first.
OpenCTI patterns	3.2	Reuse concepts	src/repository.py, src/SithAssembly/SpectreNet/, and review commands cover relationships, confidence, and hypotheses.	Test the local schema against a deduplication and conflict corpus.
Temporal	2.8	Plan a PoC	src/SithAssembly/AssemblyCore/agent_controller.py represents short stages; no long resumable workflow exists.	Define a real local long-running job with crash/resume behavior.
QUT methodology	2.8	Reuse concepts	src/pattern_engine.py and src/SithAssembly/SignalForge/ are the analysis boundaries.	Use a synthetic corpus, false-positive metrics, and explainability.
ContextForge	2.4	Defer	src/module_runtime.py and config/module_registry.json provide local discovery.	Benchmark one gateway use case only after multiple external services exist.
agentgateway	2.2	Defer	src/server.py and the command core cover current local access.	Re-evaluate when actual MCP/A2A or gRPC demand exists.
OSINTGraph	2.0	Reference	Graph and casework are already in src/SithAssembly/GhostCluster/ and src/repository.py.	Compare only data-model, import, and query ideas.
Firecracker/gVisor/Microsandbox	1.9	Evaluate later on server	deploy/ is preflight only and starts no untrusted workers.	Demonstrate the threat model, RuntimeClass, GPU/filesystem needs, and network boundaries.

Stop Rules

- No candidate replaces the command core, evidence-bound relationship generation, or human review without a demonstrated benefit.
- Every PoC needs an adapter, measurable exit criteria, and a clean removal path.
- Finalize scores only after release, license, security, and compatibility checks.
- Exclude a service that does not reduce project complexity, provide a better standard, or improve debugging.

External Components: Bounded PoC Plan

Date: 2026-09-04. This plan is deliberately analysis-first. No section starts a service, installs dependencies, or migrates data. Every PoC runs alongside the existing local workflow and must remain removable.

Phase 0: Contracts Before Infrastructure

1. Policy input: A JSON-compatible object for role, module, capability, data

class, action, and limits. Anchor: `config/agent_registry.json`. Success: five existing decisions can be evaluated reproducibly; policy errors deny access.

2. Event envelope: `schema_version`, `event_id`, `trace_id`, `occurred_at`,

`actor`, `subject`, `payload_ref`, and `idempotency_key`. Anchors: `src/SithAssembly/Conclave/agent_coordination.py` and

`src/SithAssembly/AssemblyCore/agent_controller.py`. Success: a local simulation detects a duplicate event and deterministically replays a stored sequence.

3. Provenance envelope: Run ID, module/model profile, input fingerprint,

source references, output fingerprint, and error state. Anchors: `src/SithAssembly/CipherLedger/evidence_integrity.py`,

`src/SithAssembly/EvidenceVault/evidence_vault.py`, and `src/runtime_logging.py`. Success: an artifact can be traced locally to its integrity and origin without putting plaintext evidence in telemetry.

Phase 1: Individual, Replaceable PoCs

PoC	Candidate	Scope	Exit criterion	Stop criterion
Policy	OPA	Test five fixed local access decisions through an adapter.	Allowed and denied cases are reproducible; failure is fail-closed.	No clear benefit over declarative local rules.
Event	NATS JetStream	Test one event type with two producers, three consumers, worker crash, and replay.	No duplicate case action; backpressure and recovery are measurable.	Extra operations without a robustness gain.
Workflow	Temporal	Model one long local analysis/import run with checkpoint, timeout, and manual pause.	Crash/resume preserves state and evidence references.	No representative long-running job or excessive operational burden.
Telemetry	OpenTelemetry	Create a Command -> Policy -> Worker -> Artifact trace and test redaction.	No secrets or evidence contents in exports; correlation remains traceable.	Unstable trace contract or worse debugging than JSONL.
Graph quality	OpenCTI and QUT patterns	Evaluate a synthetic corpus with aliases, conflicts, coordinated, and normal groups.	Confidence, inferred, false positives, and evidence links are traceable.	No gain over simpler rules.

Order and Guardrails

1. Write the three contracts first and protect them with unit tests.
2. Run no more than one service PoC at a time.
3. Before every PoC, verify primary sources for releases, licenses, advisories, and self-hosting.
4. Do not automate platform activity, collect data without authorization, or export case contents to external observability services.
5. Make a separate integration decision only after a PoC passes.